MODULE 24 · DOMAIN 3 - DEPLOYMENT AND ORCHESTRATION

Safe deployment

Blue/green, canary, linear, auto-rollback, shadow variants and A/B testing.

MODULE 24 of 43 DOMAIN 3 - Deploy WEIGHT 24%

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A model that passed every test can still fail on **real traffic** it has never seen.

Blue/green is the default, not an option

- **SageMaker AI automatically uses a blue/green deployment** whenever you update an endpoint.
- It provisions a new **green fleet**, shifts traffic to it, and then terminates the **blue fleet**.
- That is why endpoint updates do not cause downtime. You get this without asking.
- What you **opt into** is the traffic shifting mode and the auto-rollback alarms.
- Endpoint configurations are immutable, from module 21. That is what makes this possible.

> ON THE EXAM
>
> The distinction matters: you cannot 'enable blue/green'. You already have it. You configure how the traffic moves.

Traffic shifting modes

MODE	BEHAVIOUR, AND WHAT IT OPTIMISES FOR
All at once	100 percent shifts once green is ready, then one baking period. Fastest update
Canary	A small canary first, baking period, then the rest. Two steps. Balanced
Linear	n evenly spaced steps, with a baking period between each. Lowest risk

ON THE EXAM

AWS frames it as a trade-off: all at once minimises duration, linear minimises risk, canary balances the two.

All at once

- The green fleet becomes available, and **100 percent of traffic shifts to it**.
- CloudWatch alarms then monitor it for the **baking period**.
- If no alarm trips, SageMaker AI terminates the blue fleet. If one does, all traffic rolls back.
- **Every user is exposed** to the new model from the first second.
- Right when the update is low risk, or when the update must finish quickly.

> ON THE EXAM
>
> It is still safer than no guardrails, because the blue fleet survives the baking period and rollback is automatic.

Canary traffic shifting

- A portion of the green fleet turns on and receives traffic. That portion is the **canary**.
- **The canary must be 50 percent or less of the new fleet's capacity.** A documented limit.
- The baking period runs with **both fleets partially active** and receiving traffic.
- If no alarm trips, all remaining traffic shifts, and a **final baking period** follows.
- Any issue with the updated model **only impacts the canary**, which is the whole point.

> ON THE EXAM
>
> Two steps, so two decision points. The 50 percent cap is the detail most likely to appear as a wrong option.

Linear traffic shifting

- Traffic moves in **n linearly spaced steps**, and you choose how many.
- Each step activates part of the green fleet, deactivates part of the blue, and bakes.
- **An alarm at any step terminates the process** and returns 100 percent of traffic to blue.
- More steps means more decision points, and a longer overall deployment.
- AWS describes it as extending canary's two steps to n steps.

> ON THE EXAM
>
> Linear is the most controlled option and the slowest. If a scenario says duration does not matter, that is permission to use it.

Canary or linear?

CANARY, WHEN

Two steps is enough

- One validation point before full exposure.
- The update should not take all day.
- Canary capped at 50 percent of the fleet.
- A balance of risk and duration.

LINEAR, WHEN

You want many checkpoints

- Risk matters more than speed.
- Gradual exposure, step by step.
- You choose the number of steps.
- The most controlled option available.

THE TIEBREAKER

Both roll everything back on any alarm. The difference is how much traffic saw the new model before that happened.

The baking period

- A set amount of time during which CloudWatch alarms monitor the green fleet.
- **If any alarm trips, all endpoint traffic rolls back to the blue fleet.** Immediately.
- Set with **WaitIntervalInSeconds**, which is the interval between traffic shifts.
- A **final baking period** runs after all traffic has shifted, before blue is terminated.
- Too short and a problem has not surfaced yet. Too long and every deployment takes hours.

> ON THE EXAM
>
> Size the baking period to how long a problem takes to appear. Error rates show in seconds; quality problems take longer.

Auto-rollback, and its prerequisite

- You specify CloudWatch alarms that SageMaker AI uses to monitor the green fleet.
- **Creating those alarms is a documented prerequisite.** You must do it before the deployment.
- **No alarms means nothing can trip**, so auto-rollback cannot fire. The guardrail is decorative.
- Alarm on what would actually indicate a problem: error rates, latency, invocation failures.
- A rollback returns 100 percent of traffic to blue and terminates the shifting process.

ON THE EXAM

The traffic shifting mode controls exposure. The alarms are what actually detects a problem. Configuring one without the other is half a guardrail.

Parameters worth recognising

PARAMETER	WHAT IT CONTROLS
`WaitIntervalInSeconds`	The baking period between each traffic shift
`TerminationWaitInSeconds`	Wait after green is fully active, before terminating blue
`MaximumExecutionTimeoutInSeconds`	A ceiling on the whole deployment
`CanarySize`	How much of the fleet the canary is. Capped at 50 percent
`LinearStepSize`	The size of each step in linear shifting
`AlarmName list`	Which CloudWatch alarms trigger auto-rollback

ON THE EXAM

TerminationWaitInSeconds is the underrated one: it keeps the blue fleet alive briefly after the cutover, so rollback stays cheap.

What deployment guardrails do not cover

- The documentation carries an **exclusions section**. Not every feature supports every mode.
- They protect an **endpoint update**. They do not protect the first deployment of a new endpoint.
- They cannot detect a model being **subtly wrong**, only one that errors or slows.
- They cover minutes. **Drift over weeks** is module 29's problem, not this one.
- Check the exclusions before designing around a specific mode on an unusual endpoint type.

> ON THE EXAM
>
> A guardrail is a tripwire during a window. Everything outside that window needs monitoring, which is a different chapter.

Automating a safe deployment

- A pipeline deploys from an **approved model package** in the registry, from module 21.
- The traffic shifting mode and the alarm list are **configuration**, so they are version controlled.
- That makes safe deployment the default path rather than something someone remembers to do.
- A manual console deployment can skip all of it, which is how untested models reach production.
- Module 28 covers the pipeline. This is the deployment step inside it.

> ON THE EXAM
>
> Guardrails you have to remember are guardrails you will skip under deadline pressure. In a pipeline, they are simply how deployment works.

Traffic shifting limits **how many** users see a problem. Shadow testing means **none of them do**.

Shadow variants

- A **shadow variant** receives a copy of the traffic going to the production variant.
- **Only the production variant's predictions are returned to users.** Shadow output is logged.
- Shadow variants run on **separate instances**, so there is **no performance impact** on production.
- You compare both models' behaviour on **exactly the same data**.
- It costs the instances the shadow runs on, for the duration of the test.

> ON THE EXAM
>
> Same inputs, both models, zero user exposure. It is the only technique here that tests on real traffic with no risk at all.

Questions shadow testing can settle

- Does the new model handle the **real input distribution**, including inputs nobody thought to test?
- Does it **error** on anything? Serialisation, missing fields, unexpected values.
- What is its **latency** under genuine production load and concurrency?
- **How often do the two models disagree**, and on what kind of input?
- It cannot tell you which model was **right**, unless ground truth arrives later.

> ON THE EXAM
>
> The disagreement rate is the most useful output. A new model agreeing 99.7 percent of the time is a very different risk from one agreeing 80 percent.

Analysing a shadow test

- Start with **errors**. Any request the shadow failed on is a bug, regardless of predictions.
- Then **latency**, at the percentiles you care about rather than the mean.
- Then **disagreements**: sample them and look at what kind of input causes them.
- A high disagreement rate is not automatically bad. The new model may be right.
- Where ground truth arrives later, score both models on it once it does.

> ON THE EXAM
>
> Disagreements are where the information is. Agreeing cases tell you nothing you did not already know.

Shadow or canary?

SHADOW, WHEN

No user may be affected

- Zero customer exposure is required.
- You want behaviour on real traffic first.
- Separate instances, so production is untouched.
- Nobody receives the new model's output.

CANARY, WHEN

You are ready to deploy

- You intend to roll the model out.
- Limited exposure is acceptable.
- Real users receive the new predictions.
- Auto-rollback protects the rest.

THE TIEBREAKER

They are sequential, not alternatives. Shadow first to learn, then canary to deploy what you learned about.

A/B testing with production variants

- Two or more **production variants** behind one endpoint, with traffic **weights**.
- **Both sets of predictions reach real users**, which is what makes it a test of outcomes.
- You compare **business metrics**, not just model metrics: conversion, click-through, revenue.
- Per the ML Lens: use it to validate that a new model **performs better than the existing one**.
- It requires enough traffic to reach a conclusion, and enough time. Module 17's sample-size point.

> ON THE EXAM
>
> Shadow asks 'would this work'. Canary asks 'is this safe'. A/B asks 'is this better'. Three genuinely different questions.

Production variants and traffic weights

CONCEPT	WHAT IT DOES
Production variant	One model plus its instance configuration, behind a shared endpoint
InitialVariantWeight	The proportion of traffic each variant receives
Changing weights	UpdateEndpointWeightsAndCapacities, without redeploying
TargetVariant	Invoke a specific variant directly, for testing
Per-variant metrics	CloudWatch reports each variant separately

ON THE EXAM

Weights can be changed without a deployment, which makes A/B a dial rather than a release. That is its real advantage.

Which technique

WHAT THE SCENARIO SAYS	THE ANSWER
No customer may see the new model's output	Shadow testing
We need to know if it errors on real traffic	Shadow testing
Deploy safely, minimum risk, time is available	Linear traffic shifting
Deploy safely, but it must not take all day	Canary traffic shifting
Low-risk update, finish quickly	All at once, with alarms
Does the new model make us more money?	A/B testing with production variants
Roll back automatically if something breaks	Auto-rollback, with CloudWatch alarms configured

ON THE EXAM

Three questions, three techniques: would it work (shadow), is it safe to roll out (canary or linear), is it better (A/B).

Choosing the alarms

- **Invocation4XXErrors and 5XXErrors**: the model is failing, not merely performing poorly.
- **ModelLatency**: the new model is slower than the old one, possibly unacceptably.
- **OverheadLatency**: something in the serving path changed.
- **A custom business metric**, if your application emits one. That is the strongest signal available.
- Accuracy is usually **not** available in real time, because ground truth arrives later.

> ON THE EXAM
>
> The last bullet is the hard constraint: deployment guardrails protect against errors and latency, not against a model being subtly wrong.

Rolling back later

- Once the blue fleet is terminated, **auto-rollback is no longer available**.
- After that, rolling back means deploying the **previous model version** as a new update.
- Which is why the **Model Registry** matters: the old version is still there, approved and deployable.
- Keep the previous endpoint configuration, so the rollback is a known-good deployment.
- A subtle quality problem found a week later is always this path, never auto-rollback.

ON THE EXAM

Deployment guardrails cover minutes. The Model Registry covers everything after that, which is most of a model's life.

Practising the rollback

- A rollback path nobody has exercised is a plan, not a capability.
- Deploy the previous version deliberately, once, and time how long it takes.
- That number is what you will quote to somebody senior during an incident. Know it in advance.
- Confirm the previous model package is still **approved and present** in the registry.
- Retention policies that delete old model versions quietly remove your rollback path.

> ON THE EXAM
>
> The last bullet is worth checking today: a tidy-up job that prunes old model versions has removed the thing you roll back to.

What safe deployment costs

TECHNIQUE	COST SHAPE
Blue/green itself	Both fleets run during the shift. Briefly double capacity
Longer baking periods	Double capacity for longer. Linear costs most of the three
Shadow testing	Separate instances for the shadow, for the whole test period
A/B testing	Both variants run continuously, for as long as the test needs
Doing none of it	Free, until the day it is not

ON THE EXAM

Safe deployment costs double capacity for a while. That is the premium, and it is small against an outage.

Symptoms, and what they usually mean

SYMPTOM	USUAL CAUSE
Bad model deployed, no rollback happened	No CloudWatch alarms configured. Nothing could trip
Deployment rejected on canary configuration	Canary size above 50 percent of the fleet
Rollback fired on a healthy deployment	Alarm threshold too tight for normal variance
Deployment takes hours	Linear with many steps and long baking periods
Cannot roll back a week later	Blue fleet long terminated. Redeploy from the registry
Shadow test showed nothing useful	Too short, or no disagreement analysis done
A/B test inconclusive	Not enough traffic or time. Module 17's sample size point

ON THE EXAM

Row one is the important one. A traffic shifting mode with no alarms limits exposure and cannot detect anything.

Decide what would make you **roll back**, and alarm on that, before you deploy.

A bank is replacing the model behind its credit decisioning API. The change is substantial: a new algorithm and a new feature set. An incorrect decision has regulatory consequences.

There is no deadline pressure, and the team is content for the deployment to take several hours if that reduces risk.

Which traffic shifting mode should they use?

A All at once, with a long baking period and tight alarms

B Linear traffic shifting, with many small steps and baking periods between them

C Canary traffic shifting, with the canary set to 50 percent

D No traffic shifting mode. Deploy and monitor manually

◆ PAUSE AND DECIDE

Which traffic shifting mode should they use?

A. All at once, with a long baking period and tight alarms

Exposes every customer from the first second. A long baking period does not help anyone affected in the first minute.

B. Linear traffic shifting, with many small steps and baking periods between them

AWS describes linear as minimising the risk of introducing a new model. High-risk change, regulatory consequences, and no deadline: every condition points here.

C. Canary traffic shifting, with the canary set to 50 percent

Valid, and it exposes half the fleet's capacity at the first step. On a regulated decision that is a large blast radius for a first look.

D. No traffic shifting mode. Deploy and monitor manually

Blue/green still applies automatically, and manual monitoring gives up auto-rollback entirely. It is strictly worse than any configured option.

THE SIGNAL

'No deadline pressure' is the permission slip for linear. Without it, canary's balance of risk and duration would be the answer.

A team has a new fraud model that performed well in offline evaluation. Before any customer transaction is scored by it, they want to know how it behaves on genuine production traffic, including latency under real concurrency and whether it errors on any real input.

Compliance will not permit any customer to receive a decision from an unvalidated model.

What should they set up?

A. Canary traffic shifting with a very small canary, around 5 percent

B. A/B testing with a 95/5 traffic weight split

C. Linear traffic shifting with ten small steps

D. A shadow variant receiving a copy of production traffic

◆ PAUSE AND DECIDE

What should they set up?

A. Canary traffic shifting with a very small canary, around 5 percent

Small is not zero. Five percent of customers would receive decisions from the new model, which compliance has ruled out.

B. A/B testing with a 95/5 traffic weight split

Both variants serve real users by design. That is the whole point of A/B, and it is exactly what is forbidden here.

C. Linear traffic shifting with ten small steps

Gradual exposure is still exposure. The first step sends real customers to an unvalidated model.

D. A shadow variant receiving a copy of production traffic

Shadow output is logged, never returned to users, and runs on separate instances. It answers the latency and error questions with zero customer exposure.

THE SIGNAL

'No customer may receive a decision from it' eliminates every traffic-shifting option at once, however small the percentage.

A team configures a canary deployment for an endpoint with a four-instance green fleet. They set the canary size to three instances, reasoning that a larger canary validates the model faster.

The deployment is rejected before any traffic shifts.

Why?

A The canary size must be 50 percent or less of the new fleet's capacity

B Canary deployments require a minimum of five instances in the green fleet

C The canary must be exactly one instance, regardless of fleet size

D Canary mode is not supported when auto-rollback alarms are configured

◆ PAUSE AND DECIDE

Why?

A The canary size must be 50 percent or less of the new fleet's capacity

Three of four instances is 75 percent, above the documented cap. Two instances would be the maximum permitted here.

B Canary deployments require a minimum of five instances in the green fleet

No such minimum exists. Canary works on small fleets, subject to the 50 percent rule.

C The canary must be exactly one instance, regardless of fleet size

The canary size is configurable. It is capped as a proportion, not fixed at one.

D Canary mode is not supported when auto-rollback alarms are configured

Backwards. Alarms are a prerequisite for the deployment, not an incompatibility with it.

THE SIGNAL

The reasoning behind the mistake is sound: a bigger canary does validate faster. The cap exists precisely to stop that trade being made.

A team deploys a new model using canary traffic shifting with a 20 percent canary and a 30-minute baking period. The new model has a serialisation bug and returns HTTP 500 for about a third of requests.

The deployment completes successfully. All traffic is now on the broken model, and the blue fleet has been terminated.

What went wrong?

A The baking period was too short to detect the problem

B The canary was too small at 20 percent, so the errors went unnoticed

C No CloudWatch alarms were configured, so nothing could trigger auto-rollback

D Canary mode does not support automatic rollback. Only linear mode does

◆ PAUSE AND DECIDE

What went wrong?

A The baking period was too short to detect the problem

A third of requests failing shows in seconds. Thirty minutes is ample, and a longer period changes nothing without an alarm watching.

B The canary was too small at 20 percent, so the errors went unnoticed

20 percent of traffic returning 500s is a very visible signal. The problem is that nothing was configured to notice it.

C No CloudWatch alarms were configured, so nothing could trigger auto-rollback

The deployment completing successfully is the clue. Auto-rollback fires when an alarm trips, and with no alarms nothing can trip.

D Canary mode does not support automatic rollback. Only linear mode does

Factually wrong. All three traffic shifting modes support auto-rollback, and alarms are a prerequisite for all of them.

THE SIGNAL

'The deployment completed successfully' is the tell. The guardrail did exactly what it was configured to do, which was nothing.

What to take from this module

- **Blue/green is automatic.** You choose the traffic shifting mode and the alarms.
- **The canary must be 50 percent or less** of the new fleet's capacity.
- **No alarms means no auto-rollback.** The alarms are a prerequisite, not an extra.
- **All at once for speed, linear for safety, canary for balance.** AWS frames it exactly that way.
- **Shadow asks would it work. Canary asks is it safe. A/B asks is it better.**

ON THE EXAM

Module 25 covers deploying foundation models, where the deployment question changes shape entirely.